

Transport measurement with a horizontal and a vertical pipe microphone in a mountain stream: taking account of particle saltation

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ABSTRACT: The pipe microphone has been shown to be an effective means for monitoring bedload transport in mountain streams. It is commonly installed perpendicular to the flow direction on a stable river bed, such as that of a check dam. Acoustic pulses caused by bedload collisions with the pipe are detected by a microphone. However, bedload particles saltating over the pipe remain undetected. To overcome this disadvantage, we installed a horizontal as well as a vertical pipe microphone in the Ashi-arai-dani supercritical channel located in the Hodaka mountain range, Japan. The vertical pipe was installed on the wall of the channel and the horizontal pipe was installed on the channel bed. The acoustic response of the horizontal pipe is expected to be larger than that of the vertical pipe, because the bedload concentration decreases with increasing height above the bed. However, at high amplifications, the peak pulse value from the vertical pipe is higher than that from the horizontal pipe. We explain this observation as follows: under high bedload discharge conditions, the pulses of the horizontal pipe are saturated but those of the vertical pipe are not. We proposed a ratio (R_{hv}) between the pulses detected by these sensors, and applied this ratio for calibrating the contemporaneous pulses detected by a microphone located immediately upstream of a bedload slot sampler. Indeed the R_{hv} -corrected pulses correlated well with the bedload discharge calculated from the sampler, supporting our explanation. We conclude that bedload monitoring using concomitant vertical and horizontal pipe microphones can be used to calibrate centrally located pipe microphones when the bedload concentration is approximately homogeneous laterally across the width of the channel cross-section, and thereby represent bedload discharges more accurately than with only a single pipe microphone. Copyright © 2017 John Wiley & Sons, Ltd.

KEYWORDS: bedload monitoring; Hodaka sedimentation observatory; pipe microphone; saltation; vertical microphone

Introduction

It is well known that sediment transport equations typically overpredict bedload fluxes in mountain streams, as they do not account for limited conditions of sediment supply (Yager *et al.*, 2012). Sediment supply depends on hillslope sediment production (Fujita *et al.*, 2005; Schuerch *et al.*, 2006) or on recent stream history (Gintz *et al.*, 1996; Turowski *et al.*, 2009, 2011), so it generally varies spatially and temporally. To mitigate sediment-related disasters and control sediment transport throughout a watershed, it is essential to measure the bedload flux.

A plate geophone system has been developed in Switzerland that consists of a steel plate equipped with a geophone sensor underneath and mounted flush with the streambed. The standard steel plate length (according to flow direction) is 358 mm long, 496 mm wide, and 15 mm thick, with plates typically installed over the entire cross-section. These plates may be deployed on the downstream side of a check dam, or on a concrete bed sill. The steel plates are acoustically isolated from each other and from the steel structure by rubber elements; this is done to minimize recording the vibrations of neighboring plates or particles impacting the steel structure

(Rickenmann, 2017a). Moving bedload particles cause impact shocks on the plate, which are then transmitted to the geophone sensor. The geophone sensor induces a current that is proportional to the velocity of the plate displacement (Rickenmann *et al.*, 2012). An impulse is recorded whenever the voltage of the raw signal exceeds a preselected threshold value (typically 0.1 V), and the summed impulse counts are then stored. A linear relationship between impulses and bedload mass transported over a plate was found to apply within field sites including mountain streams in Switzerland, Austria, and Israel (Rickenmann *et al.*, 2014; Wyss *et al.*, 2016a). Thus, the Swiss plate geophone system has been installed at more than 20 field sites. This includes small steep streams in Switzerland, Austria and Italy, an ephemeral stream in Israel, and a few large rivers in Austria and the United States (Rickenmann, 2017b).

Another acoustic measuring technique - the Japanese pipe microphone - was recently developed for measuring bedload (Baezinger and Burch, 1990; Mizuyama *et al.*, 1996). It is used widely for bedload monitoring in many rivers in Japan (Hoshino *et al.*, 2004; Nakaya *et al.*, 2007). The Japanese pipe microphone consists of a steel pipe equipped with a

microphone inside the pipe. This pipe microphone is generally deployed on a rigid riverbed, such as a concrete sill or the spillway of a dam reservoir. This device converts the original acoustic signal, generated by the collision of a bedload particle with the pipe, to a pulse signal, where the summed pulse counts are then stored. It is easier to deploy the Japanese pipe microphone system than the Swiss plate geophone system, as the pipe can be directly fixed onto the concrete streambed without the need for a steel structure, making installation costs relatively low. However, a disadvantage of the pipe microphone system is that the pipe can be damaged by the movement of large boulders, and can be washed away during a large discharge event. Another important difference between the two systems is the detection sensitivity for small bedload particles. Specifically, the particle size detection limit is 4 mm for the Japanese pipe microphone (Mizuyama *et al.*, 2010a) and ~10 mm for the Swiss plate geophone (Wyss *et al.*, 2016b; Rickenmann, 2017a).

Bedload transport measurements using a microphone attached to a steel pipe were first reported by Taniguchi *et al.* (1992). Using flume experiments, they found that the number of pulses shows good linear correlation with bedload mass. The system was further studied in both flumes and field sites (Mizuyama *et al.*, 2010a, b). For example, Mizuyama *et al.* (2010a) undertook a series of flume experiments examining the pipe microphone system response to bedload transport using different pipe lengths, positions of particle collisions on the pipe, signal amplification levels, bedload discharges, and particle sizes. These experimental results showed that the pipe length affects the number of pulses, where a longer pipe dampens acoustic signals for high bedload fluxes. In addition, the particle collision location (at the pipe tip, center, or root) affects the number of pulses, with the number of pulses increasing towards the end with the microphone (the root end). Lowering the signal amplification level (microphone sensitivity) improves the recording of transport conditions under high bedload discharges; specifically, a half-buried pipe is practical for detecting bedload particles coarser than at least 4 mm. Previously, a field calibration study using the Japanese pipe microphone system was undertaken in several gravel bed rivers in Japan. This study monitored bedload transport rates using slot samplers deployed downstream from pipe microphones (Mizuyama *et al.*, 2010b). Results from the two gravel bed rivers show that a suitable calibration may be obtained for data averaged over every 5 min for intermediate bedload discharges (0.01–0.1 kg/s/m). However, during low bedload discharges (< 0.01 kg/s/m), the time over which data are averaged must be increased. At high bedload discharges (> 0.1 kg/s/m), the acoustic signal becomes saturated; however, by using the relationship between sensitive (amplification level 256 or 1024) and less sensitive (amplification level 1 or 4) channels, a calibration relationship can be obtained even when the acoustic signal of sensitive channels becomes saturated during high bedload discharge. Recently, a new method to estimate bedload discharge from the acoustic signal of the Japanese pipe microphone system was proposed (Suzuki *et al.*, 2010; Uchida *et al.*, 2013). This method is based on the proportional relationship between the acoustic signal amplitude and the product of bedload discharge and detection rate (which is the ratio between detected bedload particles and the number of bedload particles colliding with the pipe). Previous studies experimentally confirmed that the detection rate decreases with increasing particle collisions, due to the superposition of acoustic signals. To obtain the relationship between detection rate and the number of particle collisions, these studies used Monte Carlo simulations rather than experimental approaches. These studies also compared the

accuracy of this estimation approach to the direct measurement of bedload within a gravel bed river (Uchida *et al.*, 2013). These results show that estimated bedload discharge agrees well with bedload discharge measured by a direct sampling system (Ura *et al.*, 2002). This study also applied this estimation method to ~20 pipe microphone systems within streams in Japan, and showed that general trends in the estimated sediment discharge agree with deposition rates measured at sediment control dams downstream of the pipe microphone system. Another experimental and theoretical approach has been used to estimate bedload discharge from pipe microphone measurements (Goto *et al.*, 2014). This study confirmed a linear relationship between bedload particle moment and the maximum amplitude of the acoustic signal generated by the collision of a particle with the pipe. They estimated bedload discharge as the product of particle volume obtained by the above relationship and the number of particles measured by the pipe microphone.

Several studies outside Japan also deployed the pipe microphone within both field and flume conditions. Bedload transport was monitored using the Japanese pipe microphone system to elucidate the temporal variability of bedload transport in an Alpine catchment in the Italian Alps (Mao *et al.*, 2014). The results indicated that runoff generation processes (including snowmelt, glacier melt, and rainfall events) may play an important role in determining sediment supply and temporal availability in glacial mountain streams. In another study, a calibration relationship between pulses and bedload transport rate, together with the prediction of bedload particle size, was proposed by accounting for the relationship between pulses registered by different amplification levels (Mao *et al.*, 2016). Flume-based data were calibrated for grain size and for the intensity of transported sediments coarser than 9 mm.

Bedload transport was also monitored using several other impact plate systems (Downing, 2010; Krein *et al.*, 2016) and underwater microphones (hydrophones) (Geay *et al.*, 2017). Several comprehensive reviews of these techniques and devices have been published, for example, two studies by Rickenmann (2017a, b).

Although some studies estimated bedload discharge only by surrogate methods (Suzuki *et al.*, 2010; Uchida *et al.*, 2013; Goto *et al.*, 2014), it is necessary to monitor the bedload discharge by a direct method, so that a calibration relationship between the acoustic signal and the bedload discharge can be obtained, or to verify a proposed signal interpretation method. We installed a pipe microphone and a bedload slot sampler (Laronne *et al.*, 2003) in the Ashi-arai-dani supercritical channel located in the Hodaka mountain range, Japan. The weight of trapped bedload is measured by load cells installed at the bottom of the bedload sampler, and the bedload discharge is obtained from the weight change and correlated with the output from the pipe microphone system (Tsutsumi *et al.*, 2013). Although the pipe microphone has been used by authorities for bedload monitoring in mountain rivers in Japan throughout the last 10 years, some problems remain to be solved. One disadvantage of the pipe microphone system is pulse saturation caused by simultaneous collisions of bedload particles: the number of pulses is underestimated due to the overlapping of acoustic signals during high bedload discharge conditions. Employing a mean acoustic wave level (acoustic energy) instead of a number of pulses has been proposed to solve this problem of pulse saturation (Mizuyama *et al.*, 2008, 2011). The analytical method proposed by Suzuki *et al.* (2010) originally takes account of pulse saturation by considering the interference of acoustic waves.

Another disadvantage of bedload monitoring by pipe microphone is missing bedload collisions with the pipe due to the

saltation of particles. When monitoring by both pipe microphone and a direct bedload sampling method, the direct correlation between pulses and bedload discharge implicitly accounts for this reduction of bedload detection due to particle saltation. However, because particle saltation behavior changes depending on hydraulic conditions (e.g. water depth and flow velocity) and bedload particle size, correlation curves between the pulses and the bedload discharge obtained for multiple flood events may not be consistent because of different hydraulic conditions and bedload size distributions (Mizuyama *et al.*, 2010b; Tsutsumi *et al.*, 2013). Furthermore, because the analytical method of Suzuki *et al.* (2010) does not consider the reduction of bedload due to particle saltation, bedload discharge tends to be underestimated.

In this study, we propose a novel measurement method using a combination of a vertical and a horizontal pipe microphone, to confront issues in bedload monitoring methods by the pipe microphone system. We assumed that a vertical pipe microphone could detect bedload particles that saltate over the horizontal pipe on the bed, such that a larger proportion of bedload particles that pass through the cross-section formed by the horizontal and vertical pipes could be detected. This

set-up may enable more accurate measurement of bedload discharge, producing a more robust correlation between pulses and bedload discharge obtained by a direct sampler.

Measurement methods and study site

The Japanese pipe microphone system used with this study consisted of a steel pipe with a microphone unit, a converter, a pulse input unit, and a data logger (Figure 1). The pipe had a diameter of 48 mm, a wall thickness of 2 or 3 mm, and a length of 0.5–2.0 m. It was generally deployed across a rigid riverbed, such as a concrete channel or the spillway of a dam reservoir. An acoustic signal processing flow chart is shown in Figure 2. This signal was created when a microphone installed inside the pipe detected air vibrations generated by the collision of a bedload particle with the pipe. These signals were then filtered and amplified, and transmitted to the converter, which again filtered and amplified the acoustic signal and transformed it into an envelope curve (100 kHz sampling frequency). This envelope curve was then amplified at six different levels (1, 4, 16, 64, 256, and 1024) and transformed

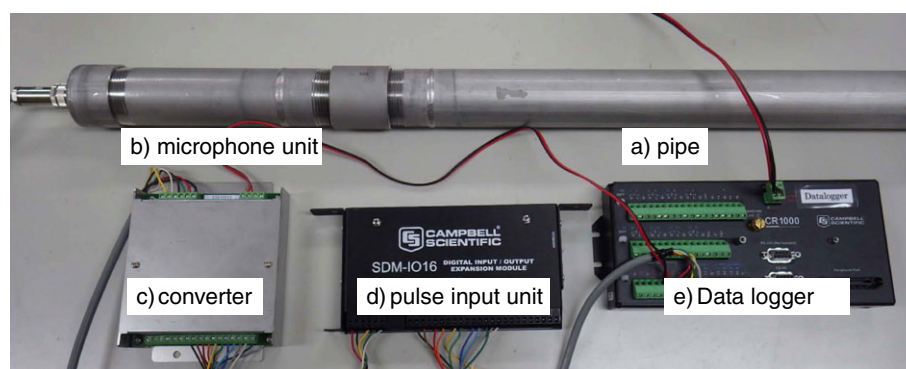


Figure 1. Devices of the Japanese pipe microphone system; (a) pipe, (b) microphone unit, (c) converter, (d) pulse input unit, and (e) data logger. [Colour figure can be viewed at wileyonlinelibrary.com]

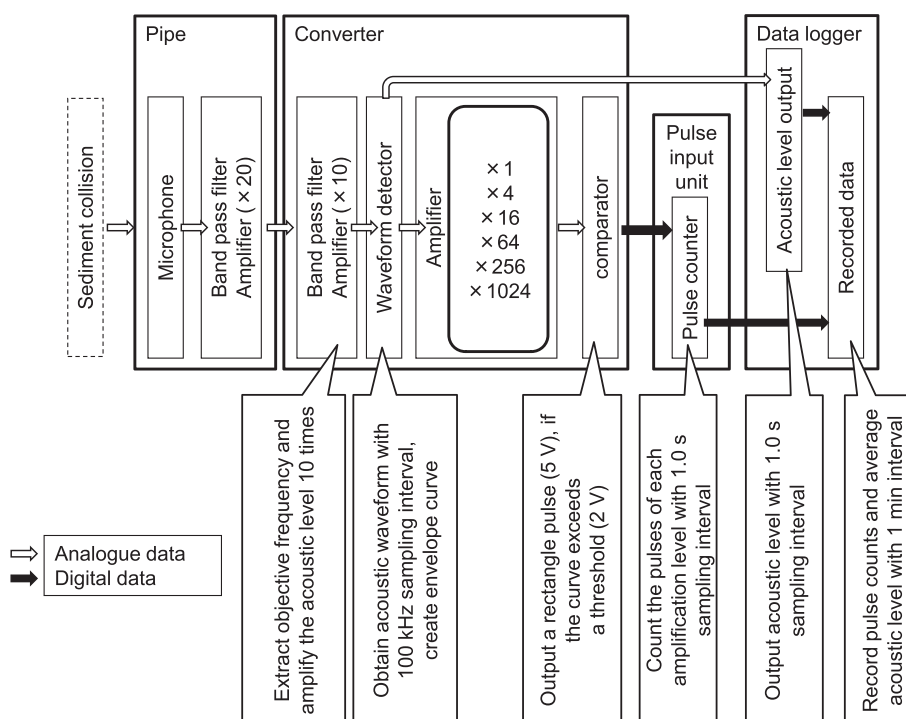


Figure 2. Flow chart for acoustic signal processing in the Japanese pipe microphone system.

into a rectangular waveform for each level (if it exceeded a 2 V threshold. This transformation (from envelope curve to rectangle waveform) was undertaken to ensure that the collision of a single particle was recorded. The number of rectangular waveforms was counted in the pulse input unit at an interval of 1.0 s, and their sum over 1 min was recorded by the data logger. The envelope curve signal amplitude was measured in parallel with the pulse count, at a sampling frequency of 1.0 Hz; the average (taken every minute) was then recorded by the data logger and used to represent the acoustic energy. An example of an acoustic signal detected by the microphone, including its envelope curve and rectangular waveform, is shown in Figure 3.

By comparison, each individual oscillating seismic waveform that exceeded a voltage threshold was defined as an 'impulse' within the Swiss plate geophone system. For the Japanese pipe microphone system, a 'pulse' was defined as a single envelope curve transformed from the oscillating acoustic signal or a rectangular waveform. Therefore, the number of pulses was ideally identical to the number of bedload particles colliding with the pipe. However, it is difficult to separate overlapping acoustic signals caused by multiple, simultaneous collisions of bedload particles, which leads to underestimation of the bedload transport rate. An example of overlapping acoustic signals, envelope curves and rectangular waveform is shown in Figure 4. This example shows three bedload particles colliding with the pipe, where the acoustic signals of the first two particles overlap, leading to the detection of only two pulses by the system. As the bedload discharge rate increases, overlap of acoustic signals becomes more intense, leading to a decrease in the number of detected pulses (which can also drop to zero) (Mizuyama *et al.*, 2010a). This

dampening of acoustic signal within the Japanese pipe microphone system is called 'pulse saturation'. A previous study showed that the threshold bedload discharge that causes pulse saturation depends on pipe length (Oda *et al.*, 2008). Under field conditions present within the Ashi-arai-dani stream, threshold bedload discharge pulse saturation values were 0.015 kg/m/s for a 5.0 m pipe (Mizuyama *et al.*, 2008) and 0.08 kg/m/s for a 0.35 m pipe (with both 1024 and 256 amplification levels, according to recent observations).

It is known that the relationship between the acoustic amplitude and the colliding bedload particle momentum is linear for a pipe system (Goto *et al.*, 2014). Therefore, larger bedload particles create a higher amplitude acoustic pulse. The purpose of investigating amplification over six different levels in the Japanese pipe microphone system is to specify bedload particle size (Tsutsumi *et al.*, 2010). Specifically, Wyss *et al.* (2016a) successfully obtained the bedload grain size distribution using the Swiss plate geophone signal based on classifying the amplitude of the different impulses. This corresponds to different amplification factors used within the Japanese pipe microphone system.

Bedload monitoring has been conducted in the Ashi-arai-dani catchment on the northern side of Mt Yakedake in the Hida mountain range, Japan (Figure 5). The drainage area of Ashi-arai-dani catchment is 6.5 km², and its elevation ranges from 1100 to 2455 m. The annual precipitation is ~2500 mm, with a mean flow discharge of 0.3 m³/s. The dominant lithology is andesite, with a considerable area covered by a welded pyroclastic flow deposit. Because of its fragile lithology and abundant precipitation, an average of 10 bedload transport events occur every year. A typical Japanese pipe microphone (diameter = 4.8 cm, length = 35 cm) and a slot sampler (slot

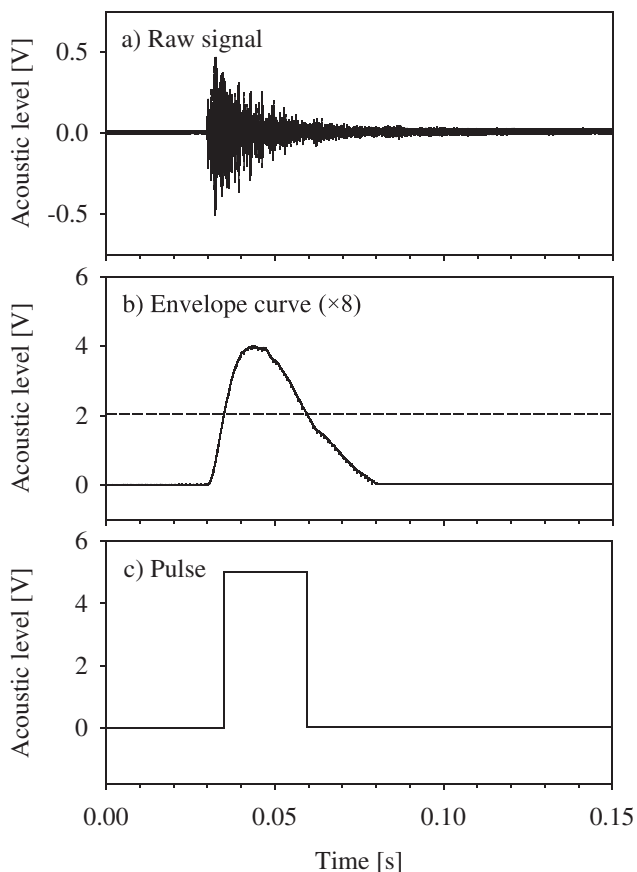


Figure 3. An example of (a) an acoustic signal detected by microphone, (b) envelope curve and (c) rectangular waveform.

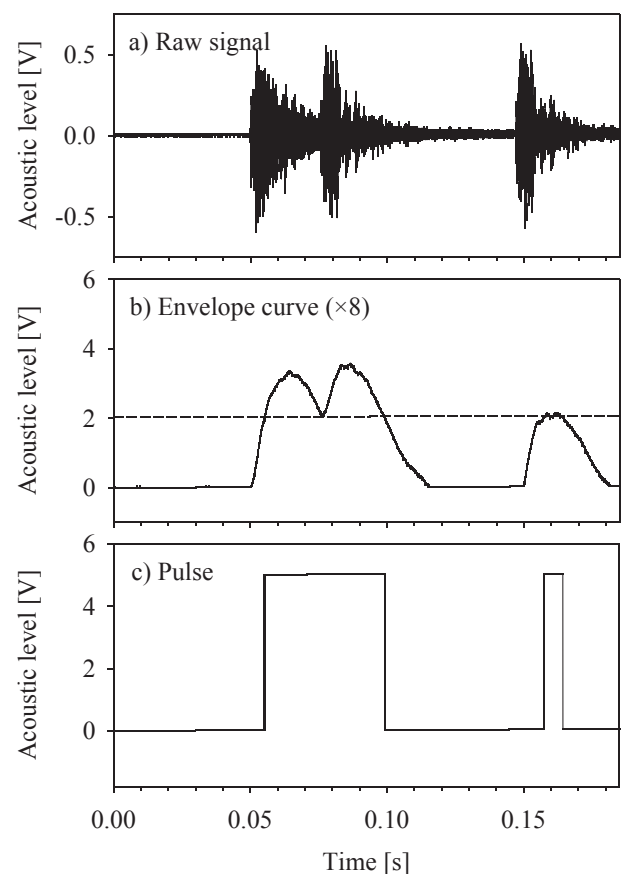


Figure 4. An example of (a) overlapping acoustic signals, (b) envelope curves and (c) rectangular waveforms.

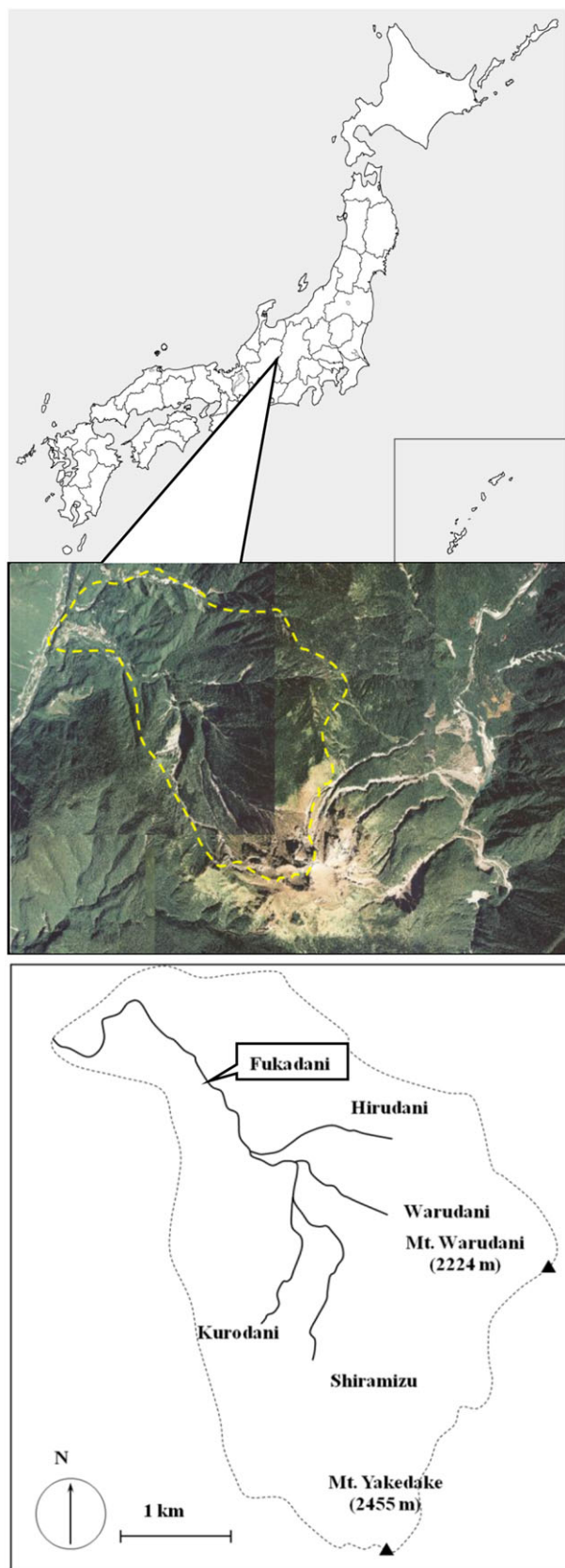


Figure 5. The bedload monitoring site of the Ashi-arai-dani catchment on the northern side of Mt Yakedake in the Hida mountain range, Japan. [Colour figure can be viewed at wileyonlinelibrary.com]

width of 20 cm) were installed in the center of the supercritical channel (length 11 m, width 5 m, slope 1/20) at the Fukadani station in the downstream portion of the Ashi-arai-dani catchment (Figure 6(a)). In addition to the central pipe, a vertical pipe

and a horizontal pipe were installed on the right side of the channel (Figure 6(a)–(c)). The vertical pipe (length 50 cm) was installed on the bank wall of the channel and the horizontal pipe (length 35 cm) was installed on the channel bed 2.0 m downstream from the vertical pipe (Figure 6(b), (c)). The pipes were both fixed to the wall and bed by half burying them and reinforcing them with concrete, leaving one-third of the pipe circumference exposed to the flow of water (Figure 6(d)). There are several ways to deploy the pipe in field sites (Mizuyama *et al.*, 2010b); these include leaving the whole circumference of the pipe exposed (which was the initial method of measuring the Ashi-arai-dani in Mizuyama *et al.*, 2004; Mao *et al.*, 2014), half-burying the pipe (Taniguchi *et al.*, 1992; Mao *et al.*, 2016), and half-burying and reinforcing the pipe with concrete (leaving one-third of the pipe circumference exposed) (Mizuyama *et al.*, 2008; Uchida *et al.*, 2013). Recently in Japan, the approach of half-burying pipes and reinforcing them with concrete was recognized as the best way of ensuring sensor stability. Only the submerged part of the vertical pipe microphone can detect the collision of bedload particles (including saltating ones). Both flow velocity and depth are monitored in the Fukadani station channel. During the study period (April 2015), flow velocity ranged from 1.9 to 3.0 m/s, flow depth ranged from 4.0 to 16.0 cm, and the Froude number ranged from 2.2 to 3.7. Because the flow velocity ranged from ~2 to 3 m/s, the time lag between bedload detection by vertical and horizontal pipe microphones was ~1 s. Although pulses were counted at an interval of 1.0 s, their sum over 1 min was recorded by the data logger (as previously discussed). Therefore, the 1 s time lag between the vertical and horizontal pipe microphones was negligible for bedload discharge analysis. The D_{50} and D_{95} of grain size for the Ashi-arai-dani streambed surface are 100 and 600 mm, respectively. Large boulders (up to 2 m in size) exist, but do not typically move during ordinary flood events. Grain size distributions of sediment trapped in the bedload sampler (which represents the transported bedload particle size) are shown in Figure 7; specifically D_{50} ranged from 3 to 7 mm, and D_{95} ranged from 20 to 30 mm.

There are three different modes of bedload particle motion in flow fields: sliding, rolling and saltation. A previous experimental study showed that spherical particles with a diameter that ranged from 5.3 to 17.5 mm are dominantly transported in the saltation mode, where a smaller proportion moves according to rolling mode when the flume bed is smooth and the flow velocity is high, where the rolling probability decreases with increasing flow intensity (Auel *et al.*, 2014). Another experimental study indicated that nonspherical particles with a diameter from 2 to 100 mm are also transported in saltation, rolling, and sliding modes (Koshiba *et al.*, 2017). Because the channel flow conditions at the Fukadani station are similar to those previous studies, bedload particles are also likely to be transported in saltation, rolling, and sliding modes. A particle moving with rolling and sliding modes always collides with the pipe deployed across the riverbed; however, a particle moving under the saltation mode may jump over the pipe and remain undetected by the microphone. During preliminary field experiments, we supplied bedload particles of various sizes to the channel at the Fukadani station (flow depth = 55–118 mm, flow velocity = 2.1 m/s, and Froude number = 2.0–2.8) six times, and monitored pulses detected by the pipe microphone. The average number of detected pulses was 7, for particles with a diameter of 22.5 mm (10 particles), and the number of detected pulses was 140 for particles with a diameter of 5.9 mm (900 particles) (Tsutsumi *et al.*, 2015). This indicates that the detection ratio for bedload particles with a diameter of 22.5 mm was 0.7, and it was 0.15 for particles with a diameter of 5.9 mm.

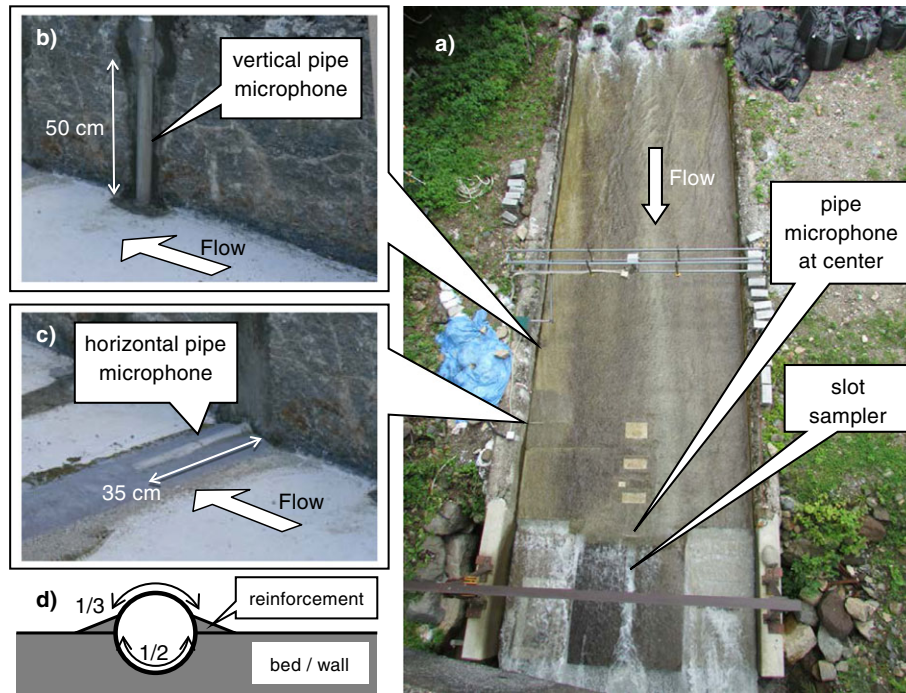


Figure 6. (a) Schematic of field pipe microphones and bedload slot sampler deployed in the supercritical channel of the Ashi-arai-dani catchment, including: (b) a vertical pipe microphone; (c) a horizontal pipe microphone; and (d) the installation of a pipe on a concrete bed or wall. [Colour figure can be viewed at wileyonlinelibrary.com]

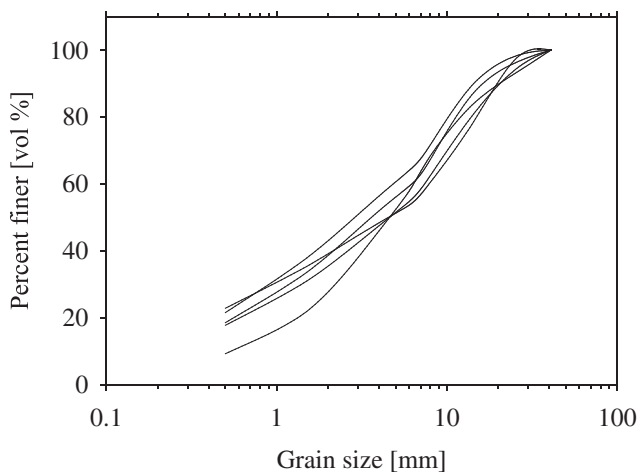


Figure 7. Grain size distribution of trapped sediment in the bedload sampler.

The validity of this detection ratio is very unclear as the number of particles supplied is small (especially for the 60 particles with a diameter of 22.5 mm); however, it is interesting that the estimated ratio was < 1.0 (and was even lower for smaller particles). Although overlapping pulses may contribute to the rather low detection ratio observed for smaller particles, bedload particles that are undetected due to saltation could also decrease the detection ratio.

Method of analysis

In this section, we propose a new method for bedload analysis that can take account of the particles saltated over the horizontal pipe microphone. Assuming a homogeneous distribution of bedload particles within a cross-section (Figure 8(a)), the pulses

detected by the horizontal and vertical pipe microphone per unit length are equal:

$$\frac{P_h}{L} = \frac{P_v}{H} \quad (1)$$

where P_h and P_v are the pulses detected by the horizontal and vertical microphones, respectively, L is the length of the horizontal microphone, and H is the water level. However, in reality, the bedload distribution is not homogeneous in vertical profile, and bedload particles are generally concentrated close to the bed (Figure 8(b)). Therefore, the pulses detected per unit length by the horizontal and vertical microphones show the following relationship:

$$\frac{P_h}{L} > \frac{P_v}{H} \quad (2)$$

Because it is impossible to detect the locations of particle collisions on the vertical pipe, we cannot determine the vertical profile of the bedload distribution shown in Figure 8(b) with the measurement by the vertical pipe microphone. However, we can estimate the reduction of bedload particles within the cross-section from the hypothetical homogeneous distribution of bedload using the ratio of pulses per unit length detected by the vertical pipe to that detected by the horizontal pipe:

$$R_{hv} = \frac{P_v}{H} / \frac{P_h}{L} \quad (3)$$

where R_{hv} is the ratio of the pulses per unit length detected by the vertical and horizontal pipe microphones. Under ideal conditions (no saturation of pulses), the ratio R_{hv} may have a value of $0.0 < R_{hv} < 1.0$, according to Equation (2). The hypothetical total number of pulses that may be caused by all bedload particles passing through the cross-section $L \times H$

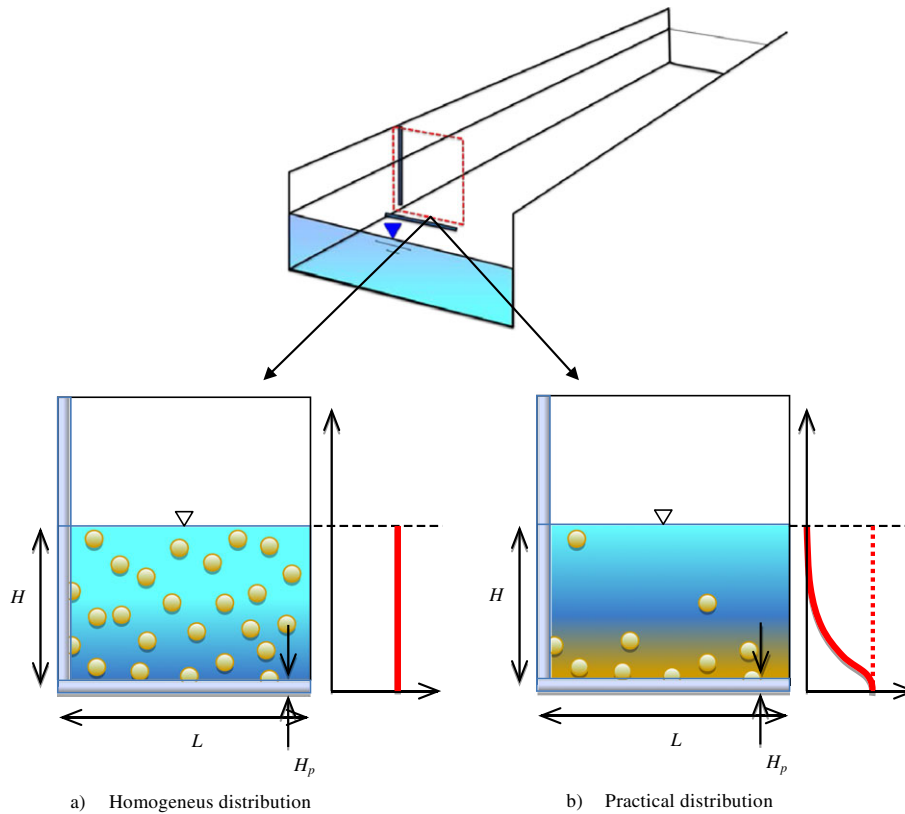


Figure 8. Observation hypothesis for the set of horizontal and vertical pipe microphones. [Colour figure can be viewed at wileyonlinelibrary.com]

under the condition of hypothetical homogeneous distribution (Figure 8(a)) is:

$$P_{t,hyp} = \frac{H}{H_p} P_h, \quad (4)$$

where $P_{t,hyp}$ is the hypothetical total pulses for a homogeneous bedload particle distribution and H_p is the height of the exposed part of the horizontal pipe, which detects the particle collisions. Therefore, the total number of pulses that may be caused by all bedload particles under the real bedload distribution conditions (Figure 8(b)) is

$$P_{t,real} = R_{hv} \frac{H}{H_p} P_h \quad (5)$$

where $P_{t,real}$ is the total pulses under the real bedload distribution conditions.

When we employ the acoustic energy (mean acoustic wave level) instead of the pulses as the bedload detection value, the ratio of acoustic energy per unit length detected by the vertical (A_v) and horizontal (A_h) pipe microphones, R_{hv-a} , and the total acoustic energy that may be caused by all bedload particles $A_{t,real}$ are as follows:

$$R_{hv-a} = \frac{A_v}{H} / \frac{A_h}{L} \quad (6)$$

$$A_{t,real} = R_{hv-a} \frac{H}{H_p} A_h \quad (7)$$

In the present research, we conducted field monitoring using a set consisting of vertical and horizontal pipe microphones, calculated the ratio R_{hv} by Equation (3) or (6), and substituted

the ratio into Equation (5) or (7) to obtain the hypothetical total values (pulses or acoustic energy) that represent the total bedload particles passing through a rectangular cross-section whose two sides are the horizontal and vertical pipe microphones. The value of P_h or A_h , in Equation (5) or (7), respectively, can be monitored by an ordinal pipe microphone installed at the center of the channel (Figure 6). In such a case, the vertical bedload particle distribution in the entire channel is approximately equal to that on the right side of the channel. Transported bedload particles are sometimes unevenly distributed in the lateral direction of a stream due to curvature in the upstream reach or distributed flow velocity. Because the upstream reach of our study site consisted of an 11-m-long straight concrete channel with a rectangular cross-section, the bedload transport may be evenly distributed in the lateral direction. Therefore, the R_{hv} ratio pertains to the entire cross-section.

Result of field observations

We monitored bedload using a set of vertical and horizontal pipe microphones in the observation channel of the Ashi-arai-dani catchment in August 2014. In April 2015, we observed a series of rainfall events (Figure 9) and several sediment transport events caused by water discharge from the rainfall events along with snowmelt. The changes of water level in the channel and the weight of the bedload sampler are shown in Figure 10 for the period 1–10 April 2015. In the figure, data for the dates 3 and 8 April are missing due to a technical problem. The water level rose and fell corresponding to rainfall on 1 April; however, it maintained a high level after the rainfall events after April 4 (Figure 10(a)). This might be explained by snowmelt accelerated by a series of rainfall events. Flow depths were less than the length of the vertical pipe microphone during the study period. In response to higher water levels (increased discharges), the weight of the bedload sampler increased, showing especially

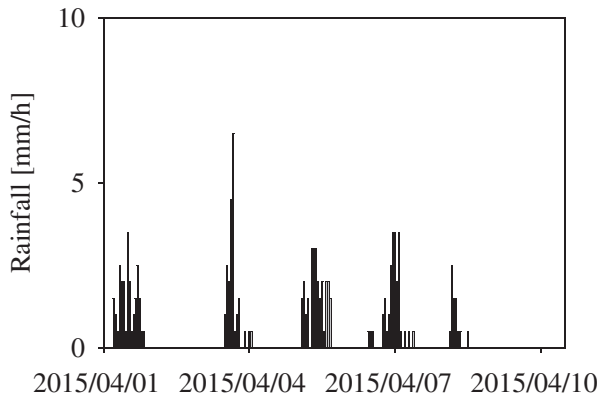


Figure 9. Precipitation during the snowmelt season (1–10 April 2015).

rapid increases on 1 and 5 April (Figure 10(b)). After 6 April, the weight of the bedload sampler remained almost constant, although the water level increased due to rainfall (Figure 10), because the pit of the bedload sampler was filled by sediment. Therefore, data from the bedload sampler are available only for the period before 5 April. Data for water level and slot sampler are available from 1 April.

The pulses and acoustic energy measured by the horizontal and vertical pipe microphones installed on the right bank of the channel during the period 1–10 April, 2015, are shown in Figure 11. In the figure, pulses measured with different amplifications (1024, 256, 64, and 16 times) and the acoustic energy are shown individually from top to bottom. In principle, more pulses are detected with higher amplification, which means that the pipe microphone can detect collisions of smaller particles. The pulses measured by the horizontal pipe microphone (Figure 11, left column) follow this principle, except for the highest amplification levels (1024 and 256).

The ratios of pulses and acoustic energy per unit length between the vertical pipe and the horizontal pipe R_{hv} and R_{hv-ar} calculated by Equations (3) and (6), are shown in Figure 12. The ratio R_{hv} should basically range from 0.0 to 1.0, as we mentioned previously; however, R_{hv} sometimes took values larger than 1.0 with amplifications of 1024 and 256.

The hypothetical total pulses and acoustic energy that may be caused by all bedload particles $P_{t,real}$ or $A_{t,real}$ calculated

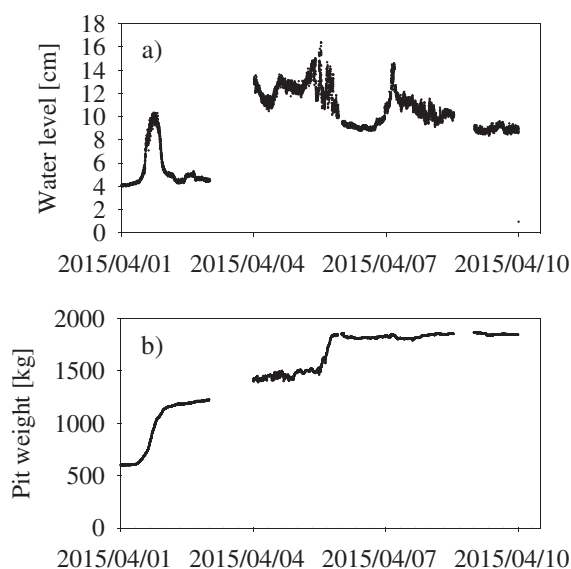


Figure 10. Changes in (a) water level and (b) bedload weight in sampler.

by Equations (5) and (7), are shown in Figure 13. In comparison with the original pulses detected by the horizontal pipe (Figure 11, left column), much higher pulse peaks for an amplification of 1024, lower pulses but sometimes higher peaks for an amplification of 256, lower pulses for amplifications of 64 and 16, and relatively similar acoustic energy were calculated. From these comparisons, we infer that estimating the total bedload is more reasonable using horizontal and vertical pipe microphones and higher amplifications of 1024 and 256 for pulse counts or the acoustic energy during low to intermediate bedload discharge events (such as the one discussed here).

To examine the appropriateness of the method for calculating the hypothetical total number of pulses or acoustic energy reflecting all the bedload particles passing through the cross-section over the horizontal pipe microphone, we calculated $P_{t,real}$ and $A_{t,real}$ for the pipe microphone at the center of the channel and examined the correlation with the bedload discharge obtained by the bedload slot sampler. The pulses measured by the horizontal pipe microphone with different amplifications and acoustic energy are shown together with the water level in the left column of Figure 14 from 06:00 on 1 April to 06:00 on 2 April, 2015. Except for the pulse with amplification of 1024, pulses decreased as amplification decreased. The change of the sediment weight in the pit of the bedload sampler and the bedload discharge are shown at the top of the right column of Figure 14. The bedload discharge was calculated as the differential of the sediment weight. To reduce the instability of the measurement of the sediment weight, a 10 min moving average was employed for the differential value. The hypothetical total number of pulses $P_{t,real}$ and acoustic energy $A_{t,real}$ reflecting all the bedload particles passing through the horizontal pipe installed at the center of the channel were calculated (or measured pulses and acoustic energy were 'corrected') with the ratio of the vertical and horizontal pipes at the right side of the channel (Figure 12) and are shown in the right column of Figure 14. As we expected, the pulses with amplifications of 64 and 16 show very low values and therefore these data are not very useful. The pulses with amplifications of 1024 and 256 and the acoustic energy were calculated to be higher than the original values shown in the left column of Figure 14, and show a similar tendency to the bedload discharge displayed at the top of the right column of Figure 14. The correlations between the bedload discharge and either the pulses or the acoustic energy measured at the center of the channel are shown in Figure 15. The left and right columns in this figure show correlations with and without correction by the ratio R_{hv} obtained by the set of vertical and horizontal pipe microphones installed at the right bank of the channel. The effect of pulse saturation can be seen in the correlation between bedload discharge and pulses with amplifications of 1024 and 256 (without correction by the ratio R_{hv} shown in Figure 15): the monitored pulses have upper limitations (especially with an amplification of 1024). The correlation between the bedload discharge and pulses corrected by the ratio R_{hv} shows an obvious linear relationship and a larger correlation line slope. This indicates that correlation with the ratio R_{hv} functions appropriately, mostly excluding the effect of pulse saturation.

Discussion

The pulses determined by the horizontal pipe microphone with higher amplifications (1024 and 256) show lower variation than those with lower amplifications (64 and 16) (left column of Figure 11). This may be related to pulse saturation, which is due to the overlap of acoustic waves caused by collisions of

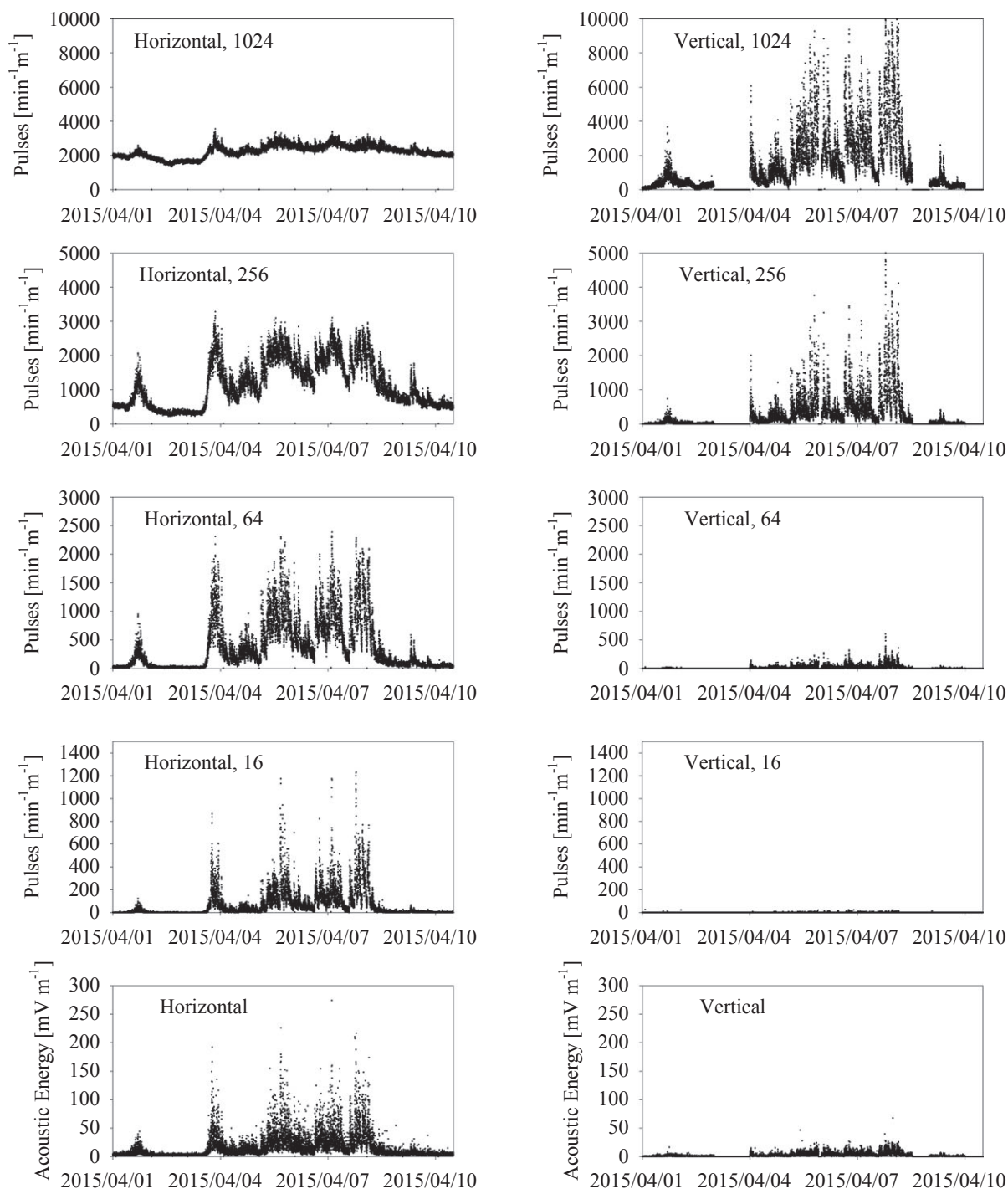


Figure 11. Observed pulses and acoustic energy by horizontal (left column) and vertical (right column) pipe microphones (measurement time resolution of 1 min).

numerous bedload particles on the highly sensitive microphone (Mizuyama *et al.*, 2008). A saturation effect was not observed for pulses measured by the vertical pipe microphone, where the number of measured pulses increases with an increase in amplification. The reason why pulse saturation did not impact measurements from the vertical pipe microphone may be that fewer particles hit the pipe. Only the lower portion of the vertical pipe captures the majority of bedload collisions (which flows mostly near the bed), whereas the whole length of the horizontal pipe acts as a sensor.

Observed pulses with amplifications of 64 and 16 measured by the horizontal pipe are greater than those measured by the vertical pipe; however, peak observed pulses with amplifications of 256 and 1024 measured by the horizontal pipe are less than those measured by the vertical pipe. This supports the

inference that pulse saturation affected pulses measured by the horizontal pipe at higher amplifications of 1024 and 256. In addition, the acoustic energy measured by the horizontal pipe is more than that measured by the vertical pipe.

The R_{hv} ratio can have values larger than 1.0 with amplifications of 1024 and 256 (as shown in Figure 12). This occurred because pulses measured by the horizontal pipe microphone with higher amplification do not reflect actual particle collisions due to pulse saturation (i.e. the right-hand side denominator of Equation (3) became smaller). The R_{hv} ratios of pulses with lower amplification and R_{hv-a} of the acoustic energy ranged from 0.0 to 1.0.

The effect of pulse saturation cannot be seen even with higher amplifications upon correcting for R_{hv} , which includes a reduction in pulse detection by the horizontal pipe

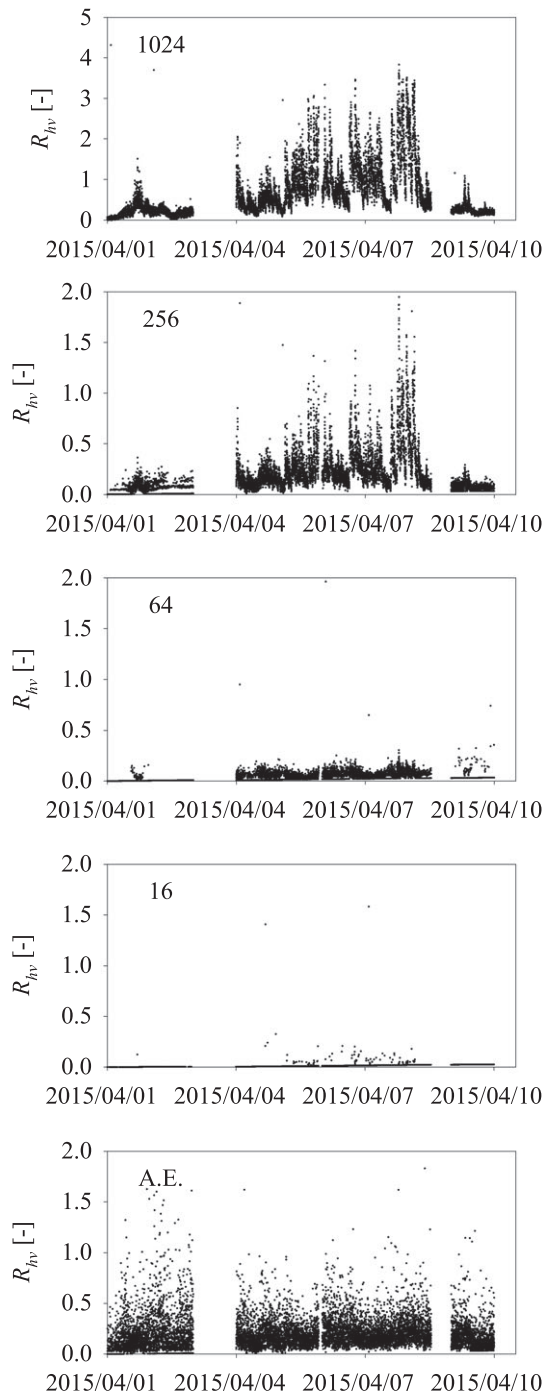


Figure 12. Measurement ratio by horizontal and vertical pipe microphones.

(Figure 13). The calculated total pulses with lower amplifications of 64 and 16 are smaller than the number of pulses measured by the horizontal pipe microphone. This may be because lower amplifications are useful for detecting larger bedload particles, where pulses measured by the vertical pipe microphone were too few to reflect an appropriate vertical bedload distribution profile.

The correlation between bedload discharge and acoustic energy without correction originally showed a linear relationship because the acoustic energy partly excludes the effect of pulse saturation, as we discussed previously (Figure 15). However, the correlation between bedload and acoustic energy with the correction shows a linear relation with a larger slope, although it shows wider scattering than that without the correction. Another correlation effect that takes particle saltating over the pipe into account contributes to this larger slope. Wider

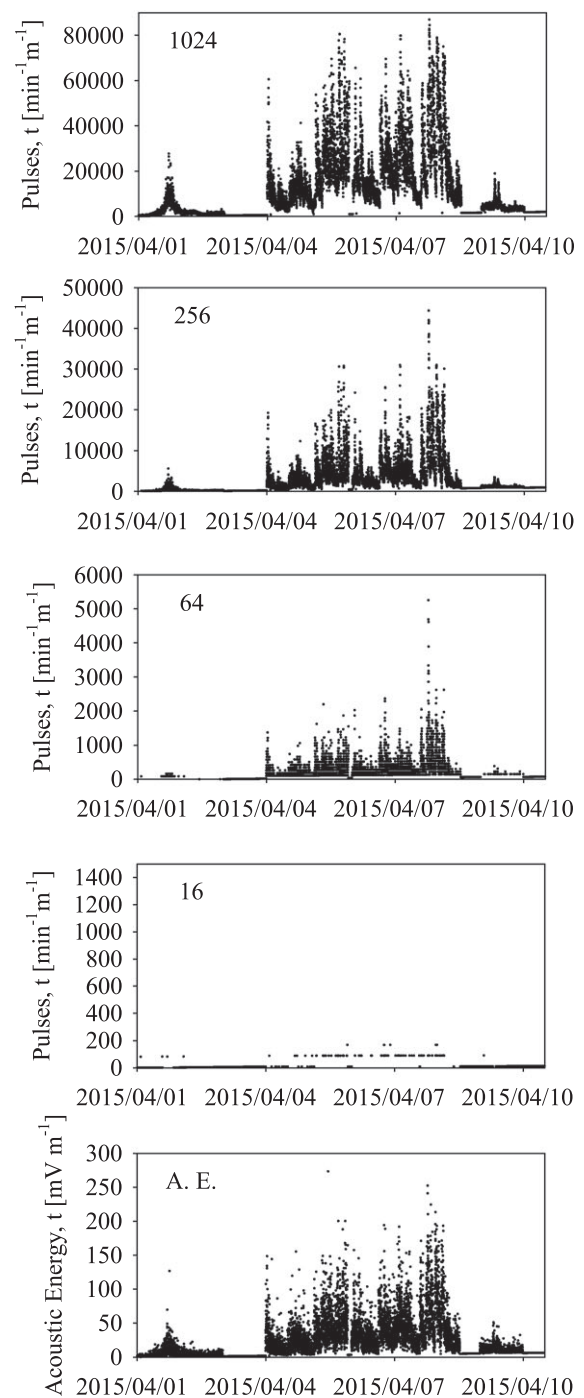


Figure 13. Hypothetical total detection by the horizontal pipe microphone, corrected by vertical measurement.

scattering may indicate a limitation of the assumption that the vertical bedload particle distribution in the entire channel is approximately equal to that of the right side of the channel under low to intermediate bedload discharge conditions. Correcting the acoustic energy is less important than it is for pulse counts from results for acoustic energy. However, the overlap of acoustic signals due to multiple particle collisions with the pipe still remains, as the synthesis of multiple oscillating acoustic signals does not always increase the acoustic level. Although the difference in calibration relationships between the uncorrected and corrected acoustic energy is small (Figure 15), there is indeed a difference in slope values. Therefore, the correction in acoustic energy will also be effective during a high bedload discharge event.

We divided the correlation between the bedload discharge and uncorrected pulses with an amplification of 1024 (top left

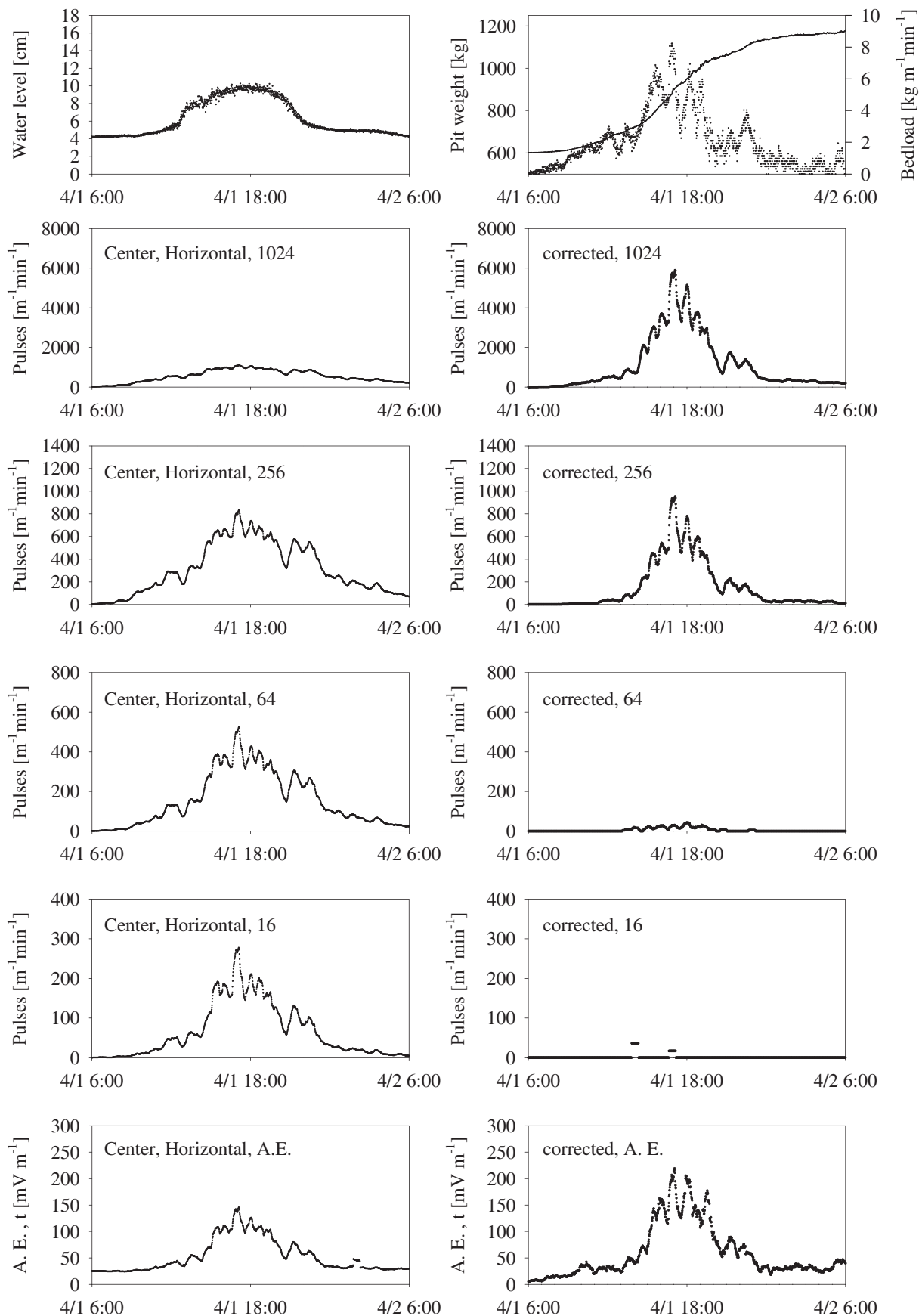


Figure 14. Changes in water level, bedload discharge, and measured and corrected pulses (acoustic energy) on 1–2 April 2015.

in Figure 15) into an unsaturated part (lower bedload discharge) and a saturated part (higher bedload discharge) depending on their slopes. The boundary bedload discharge value between saturated and unsaturated parts was estimated to be 3.2 kg/m/min (0.053 kg/m/s). This value is more than three

times larger than 0.015 kg/m/s reported by Mizuyama *et al.* (2008), and comparable with 0.08 kg/m/s recently observed at the Ashi-arai-dani station. This study used a pipe length of 0.35 m, whereas Mizuyama *et al.* (2008) used a pipe length of 5.0 m, where bedload discharge (that causes pulse saturation)

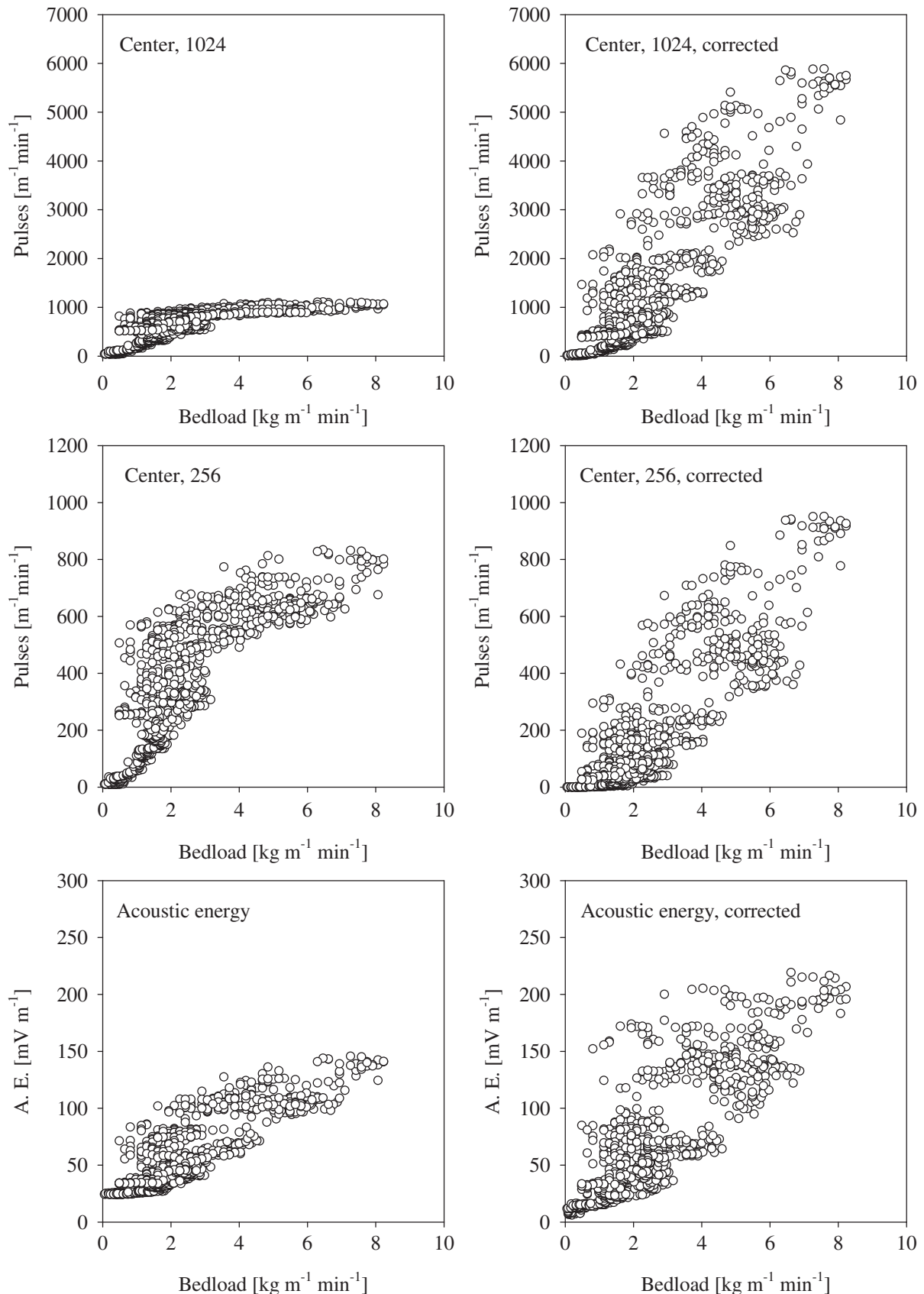


Figure 15. Correlation between bedload discharge, pulses, and acoustic energy (left column: corresponds to measurement from the horizontal pipe microphone at channel center, and the right column to corrected measurements from the vertical and horizontal pipe microphones).

tends to be smaller for the longer pipe. This agrees with the experimental results of Oda *et al.* (2008), who found that a shorter pipe microphone could reduce the pulse saturation effect. This implies that pulse measurements with high

amplification (1024) do not work for intermediate to large bedload discharge under ordinary calibration. The slope of the regression line for the unsaturated mode is $\sim 300 \text{ kg}^{-1}$ (top left of Figure 15) and 120 kg^{-1} in the work of Mizuyama *et al.*

(2008). The flume-based calibration found by Mao *et al.* (2016) showed a regression line slope ranging from 100 to 300 kg⁻¹ for relatively well-sorted bedload particles (diameter = 5.6, 13.0 mm) with an amplification of 1024. A comparison between regression line slopes is shown in Table I. The regression line slope is ~610 kg⁻¹ for the correlation between bedload discharge and corrected pulses with an amplification of 1024 (top right in Figure 15). Because the regression line for the correlation between bedload discharge and uncorrected pulses is only produced for the unsaturated part, differences between the slopes of regression lines with and without correction may reflect the detection of saltating particles by the vertical pipe microphone. These differences indicate that twice the bedload (including saltating particles) is detected by our method with both vertical and horizontal pipes.

The slope of the regression line corresponds to the K_b value of the correlation between bedload discharge and impulses in the Swiss plate geophone system. These K_b values range from 5.45 to 22.9 kg⁻¹ in mountain streams (Rickenmann *et al.*, 2014). Although the signal processing differs between the Swiss plate geophone and the Japanese pipe microphone system (including differences between impulses versus pulses, geophone versus microphone, plate versus pipe, and amplification values), we simply compared these regression line slopes. The slope of the regression line for the pipe microphone system was ~10 to 50 times greater than that for the plate geophone system. Although there are some differences in the calibration between these two systems, directly comparing measurements made between the two systems in the Swiss Erlenbach channel confirmed that similar measurements of bedload transport could be obtained by the two systems (Rickenmann, 2017a, b). But the plate geophone may detect more saltating bedload particles in contrast to the pipe microphone, as the length of the sensor in the flow direction is longer for the plate (36 cm) than the pipe (5 cm) method.

From correlations between bedload discharge and pulses or the acoustic energy shown above, we may conclude that we can employ these correlations (with correction by the ratio R_{hv} of horizontal and vertical pipe measurements), to obtain bedload discharge with more consistent results. This eliminates the pulse saturation effect for higher amplifications and takes into account saltating particles over the pipe within the hypothetical total pulse count. Because these results were obtained from only one low to intermediate bedload discharge event, it is necessary to examine the suitability of this method for higher bedload discharge conditions.

In the previous section, we calculated the ratio of pulses obtained by the vertical pipe to those obtained by the horizontal pipe, even though the pulses measured by the horizontal pipe showed obvious saturation. Here, we discuss the robustness of this method. Earlier we proposed Equation (5), which calculates the hypothetical total pulses that may be caused by all bedload particles, including particles saltating over the horizontal pipe.

If we assume a cross-section formed by the horizontal and vertical pipe rectangle (Figure 8), substituting R_{hv} of Equation (3) into Equation (5), because the P_h in Equation (3) is equal to that in Equation (5), results in the following expression for P_t :

$$P_t = \left(\frac{P_v}{H} / \frac{P_h}{L} \right) \frac{H}{H_p} P_h = \frac{L}{H_p} P_v \quad (8)$$

This equation means that P_t is proportional to P_v with the coefficient (L/H_p) and is unrelated to P_h measurement. Therefore, even if the pulses measured by the horizontal pipe P_h reach saturation, the P_t calculation is not affected by horizontal pipe measurement.

In the previous section, the ratio R_{hv} calculated by the pulses obtained by the horizontal and vertical pipes on the right bank of the channel was used for correction of the pulses obtained by the horizontal pipe at the center of the channel. In this case, the P_h in Equation (3) (by the horizontal pipe at the right bank) is not equal to that in Equation (5) (by the horizontal pipe at the center of the channel). However, if the pulse saturation levels between these two horizontal pipes are similar, the effect of pulse saturation can be eliminated. Therefore, the correction of pulses can be applicable under the condition that there is not strong variability of bedload in the horizontal direction.

Reconsidering Equation (8), it is suggested that bedload measurement by a horizontal pipe microphone P_h is unnecessary (only a vertical pipe is sufficient) for the total bedload measurement. This led to the idea that installation of a single vertical pipe microphone could be used for bedload measurements. The vertical pipe configuration is similar to a gravel transport sensor developed in the United States by Downing (2010). This gravel transport sensor consists of a long, vertical, stainless steel pressure plate that is bonded to a PVDF film and mounted to an aluminum block. When gravel collides with the plate, it generates an electric charge proportional to the impact force. This gravel transport sensor can be deployed vertically on a moving bed.

The hypothetical total pulses by all bedload particles passing through the entire cross-section ($L \times H$) during bedload measurements using a single vertical pipe microphone can be calculated by Equation (8). Measurement by the single vertical pipe has the advantage that the pipe sensor is always active even though sediment may cover the bed surface, which is sometimes a problem at a measurement site under conditions of high bedload discharge and relatively low bed gradient. On the other hand, some disadvantages should be resolved before the single vertical pipe is used practically, namely shock resistance to collisions by large gravel and degradation of sensitivity due to the accumulation of wood and litter on the pipe. Measurements by a single vertical pipe are only applicable under a laterally homogeneous bedload transport distribution. Furthermore, because H_p is very small in relation to L , there may be considerable uncertainty in the calculated P_t values.

Table I. Comparison of slopes of regression line between measurements

Research	Mizuyama <i>et al.</i> , 2008	Mao <i>et al.</i> , 2016	The present study
Site	Ashi-arai-dani stream	Flume	Ashi-arai-dani stream
Pipe length	5.0 m	0.5 m	0.35 m
Pipe deployment	Full circumference exposed	Half-buried	Half-buried, reinforcement with concrete
Microphone amplification	1024	1024	1024
Particle size	3–30 mm	5.6, 13.0 mm	3–30 mm
Slope of regression line	120 kg ⁻¹	100–300 kg ⁻¹	300 kg ⁻¹
Slope of regression line, with correction	-	-	610 kg ⁻¹

This is due to variability in the probability of detecting particles along the horizontal distance H_p , and possible changes in the exposed vertical section due to abrasion of cement upstream of the pipe. Even if these disadvantages can be resolved, there are still some limitations in the flow and bedload transport conditions. Specifically, the vertical pipe can reduce the saturation effect during high bedload transport conditions because of its limited detection width (pipe diameter D) when compared with the horizontal pipe (pipe length L). However, if bedload transport increases strongly, the acoustic signal detected by the vertical pipe will become saturated as well. The threshold bedload discharge that results in acoustic pulse saturation for a vertical pipe can be roughly estimated as L/D times that detected by a horizontal pipe. To be more precise, the estimated threshold value should include the increase of detection area by saltating bedload particles; thus, the threshold value becomes smaller than previous estimations. Thus, despite its advantages, measuring bedload by a single vertical pipe microphone is very challenging and is not presently practical.

Conclusion

In the present study, a vertical pipe microphone was installed on the sidewall of a rectangular channel (in addition to the typical horizontal pipe microphone). This new bedload monitoring analytical method was proposed as it can account for bedload saltating over the horizontal pipe microphone. During the measurement, pulses measured by the horizontal pipe saturated at higher amplification levels, which affected the ratio of pulses between the vertical and horizontal pipes. As a result, although the effect of pulse saturation in the horizontal pipe measurement and consideration of saltating particles cannot be separated quantitatively, the hypothetical total bedload pulses were calculated. The correlation between hypothetical total bedload pulses (especially with amplifications of 1024 and 256) and bedload discharge showed a better linear relationship than previously found. However, because our result was obtained from only one low to intermediate bedload discharge event, it is necessary to investigate the suitability of this method under different events, including high bedload discharge conditions.

According to the new analytical method, we also proposed that measurement with only a single vertical pipe microphone could potentially be used for bedload monitoring. However, it will be necessary to solve several vertical pipe installation problems, including shock resistance to collisions from large gravel and degradation of sensitivity due to wood or litter accumulation on the pipe. This means that there is considerable uncertainty due to variation in the probability of detecting particles along the vertical distance.

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